Course Syllabus

download (https://northeastern.instructure.com/courses/244222/files/39226376?location=course_syllabus_244222&wrap=1) (https://northeastern.instructure.com/courses/244222/files/39226376/download?location=course_syllabus_244222&download_frd=1)

PHTH 6800

Causal Inference in Public Health Research

Instructor

Instructor: Louisa Smith, Assistant Professor (Department of Public Health & Health Sciences / The Roux Institute)

E-mail: l.smith@northeastern.edu

Phone: Teams

Office hours: Fridays 10-11am (Online); Wednesdays in-person by appointment; Online by appointment

TA: Shuai Yuan (yuan.shuai2@northeastern.edu)

Course Information

Description:

Exposes students to causal inference approaches, including causal diagrams and counterfactual theory. Students are also asked to draw upon their own research experiences and prior epidemiology training to evaluate public health studies. Covers how to apply the fundamental concepts of counterfactuals and causal diagrams; assess threats to validity in study designs and analysis, including confounding, selection bias, and measurement error/misclassification; evaluate the validity of a public health research study's design and analysis with respect to addressing causal questions; and critically analyze scientific literature and apply findings to clinical or policy decisions. Offers students an opportunity to think critically and rigorously about the implications of study design and analysis toward addressing public health questions.

Prerequisites:

PHTH 6202 with a minimum grade of C-

Corequisites:

None

Credit Hours:

3

Objectives / Outcomes

- Use tools such as causal diagrams and counterfactual theory to pose causal questions and evaluate causal claims in public health
- Assess the validity of a public health research study's design and analysis with respect to addressing causal questions
- Design and analyze an observational study to answer a causal question

Teaching Methods / Learning Experiences

This course's weekly meetings consist of lecture, discussion, and individual/small group work on problems. Students will complete weekly readings and interactive discussions on the readings via Perusall, as well as weekly problem sets. Ongoing work in R throughout the semester will culminate in a final project.

Grading / Evaluation

Component	Percentage
Quizzes	20%
Article presentation	20%
Participation	25%
Final project	35%

There will be intermediate check-in points for the final project, but they will not be graded; instead, they will serve to ensure you are making progress. If you are ever concerned about how you are doing in the course, please feel free to reach out to me. Likewise, I will be in contact if I am concerned about your progress.

The participation component will be based on your attendance and participation in class and office hours as well as online participation via Perusall. Attendance is not required at office hours, but for those who participate less in class discussions, it can be a good way to participate more in the course. If you are unable to attend office hours, please let me know so we can arrange an alternate time to meet.

Grades will be rounded up to the nearest percentage point.

*See University Grading Scales for UG and Grad here: **https://registrar.northeastern.edu/article/university-grading-system/ (https://registrar.northeastern.edu/article/university-grading-system/)**

Materials / Resources

Required Text:

Readings from the academic literature are posted on Canvas and Perusall. Course notes written by the instructor will also be provided.

Supplemental Readings:

Suggested readings are posted on Canvas. ***Causal Inference: What If*** *(https://miguelhernan.org/whatifbook)* is available freely online and recommended as supplemental reading.

Canvas: ***https://canvas.northeastern.edu*** **(https://canvas.northeastern.edu)**

Course Policies

Student Expectations:

Attendance

Discussion is most fruitful when the entire class is present, so attendance in person is generally expected. Illness and major life events are of course exceptions. In fact, you are expected *not* to attend class if you are sick in order to keep us all healthy. If you are unable to attend class, please let me know in advance, if possible. For prolonged absences, contact me to discuss a make-up plan.

Readings

Readings will generally be chosen from the methodological and applied causal inference literature and may be supplemented with excerpts from textbooks (open-source or available online from the Northeastern library). Articles will be uploaded to Perusall, available via Canvas. Some readings will relate to the previous class meeting and are meant for deeper understanding; others are to prepare you for the upcoming class meeting. I will leave some comments and questions in Perusall; you are encouraged to respond to these and leave questions and comments of your own. Responses will not be quantified and graded per se, but they will contribute to the participation component of your final grade. Selected articles will be chosen for student-led discussions.

Article discussions

Each student will be responsible for (co-)leading the discussion on an article in the second half of the semester. These articles will be chosen from the applied literature and will be more focused on a specific application of the topic we talked about the *previous* week.

By the Sunday night before the class in which the article is assigned, you will be responsible for leaving comments and questions in Perusall. My comments and questions on other articles will serve as a guide for what I am looking for. In class, you will jump-start the discussion with a 5-10 minute presentation. The presentation should include a brief summary of the article, any important vocabulary and notation used in the article, and connections to material previously covered in the course. You can prepare slides, write on the whiteboard, or use any other presentation format. This will be followed by a 15-25 minute discussion of the article. You will be responsible for leading the discussion, but all students are expected to participate. A rubric for the article presentations will be provided.

The articles in the syllabus with asterisks will be those for which student presentations will be assigned. Articles will be chosen during the first week of class.

Weekly problem sets (quizzes)

Short problem sets will be given using the quiz functionality on Canvas for each week's material, following the class that covers it. (Note that I am using problem set and quiz interchangeably in this course.) You will be allowed to use course resources to complete the quizzes and discuss problems with classmates, but they should represent your own work. The quizzes are for you to assess your own understanding of the material and for me to assess the class's understanding as a whole. Quizzes should be completed by 5:00 pm on the Tuesday following the class that covers the material. However, I encourage you to look at the quiz thoroughly before the week's Friday online office hours so that you are prepared to attend if necessary and ask questions about the material.

R

We will use R for some in-class exercises and problem sets. I will provide code snippets and you will be expected to modify and extend the code as necessary for the problem sets as well as the final project.

Final project

The final project will be an analysis and write-up on a causal question of your choice, using data of your choice. You will be asked to propose a question and design a study and analysis plan to answer it. You will describe the assumptions necessary to answer the question, to what extent those assumptions are expected to hold given the observable data.

The final project will consist of a 5-10 page paper (including tables and figures as necessary, along with code in an appendix) and a short presentation during the final class meeting. A rubric for the final project will be provided, and check-in points will be assigned throughout the semester.

It is acceptable to use a research question you plan to analyze for your dissertation or another project, but you should not use a research question you have already analyzed or an analysis plan you have already prepared. I am happy to discuss potential questions with you as well as data that could be analyzed.

Office hours

Everyone is encouraged to come to office hours, no matter how you are doing with the material. Online office hours will be held on Fridays to go over questions from that week's class. I expect they will serve as an optional lab/review session and a chance to get clarification about the problem set. Other office hours can be arranged by appointment to discuss final projects, review any material that is giving you trouble, and address any other issues that may arise. I am generally only in Boston on Wednesdays but can meet online other days.

Typical weekly schedule

Wednesday	**Friday**	**Sunday**	**Tuesday**
Come to class having finished the readings, discussed on Perusall, and prepared to discuss in class	Attend online office hours with questions (optional)	If you are responsible for an article next week, finish leaving Perusall comments	Complete problem set by 5pm (and be most of the way through readings)

Class Participation:

This is a participatory course and each student is expected to be an active participant in the work that occurs. Student learning in this course is directly related to the efforts made in learning the course material. Attending class and arriving to class promptly is expected of each student (and shows respect for fellow students). Students will be called on in class, and each student can expect to be called on unexpectedly several times during the semester.

Professional Conduct:

Part of the student learning experience is to practice professional behavior. This includes affording your student peers, presenters and professors the courtesy of not using computers, cell phone, pagers or text messaging during class. Non-adherence to these expectations of professional behavior will result in dismissal from class. Consistent with University policy, students must seek the course instructor's permission to record class lectures.

Make Up Policy:

See Late Work Policy

Late Work Policy:

Quizzes are due by 5 pm on the Tuesday following the class that covers the material. Exceptions may be made for illness or other events. You are expected to complete the reading before class on Wednesday and provide comments and respond to others' comments on Perusall.

If you find that you will not be in class on the day you are due to make your article presentation due to illness or another major event, we will arrange a time for you to present later.

Check-ins throughout the semester for the final project are used to make sure you are on track. If I am concerned about your progress, I will let you know. For logistical reasons, late final projects will likely require taking an incomplete in the class and finishing over the summer, as will prolonged absences.

Academic Integrity:

As health professionals and scientists-in-training, the highest level of ethical behavior is expected of all students. Plagiarism, cheating, and any form of unauthorized collaboration will not be tolerated and will be handled in accordance with University policies including the Academic Honesty and Integrity Policy. Requirements can be found at: **https://osccr.sites.northeastern.edu/academic-integrity-policy/ (https://osccr.sites.northeastern.edu/academic-integrity-policy/)**

https://osccr.sites.northeastern.edu/code-of-student-conduct/ (https://osccr.sites.northeastern.edu/code-of-student-conduct/)

- Acts of academic dishonesty will be referred to OSCCR (Office of Student Conduct and Conflict Resolution).
- Infractions will be dealt with according to Northeastern Disciplinary Process shown in http://www.northeastern.edu/osccr/academic-integrity-report-form/
- The Turnitin application may be used by the course instructor to identify and document cases of plagiarism.

Use of Generative Artificial Intelligence (AI):

Generative Artificial Intelligence (AI) refers to large learning models (LLMs) that can generate high-quality text, images, and other content based on the data they were trained on. Popular AI "bots" include ChatGPT which produces text-based responses in the form of poems, essays, articles, letters and more based on a question entered by the user. It also can provide translation and copy language style and structure. LLMs are trained to predict the most relevant sequence of words in response to a prompt. AI tools are only able to draw on the information that was input, meaning their responses reflect the biases and limitations of the material they have been trained on. It is important to understand that responses produced by generative AI tools will likely reflect consensus beliefs, including any biases and inaccuracies that inform those beliefs.

You may not use generative AI tools to produce any assignment, or any question on an assignment, in its entirety. The language we use in this course is very precise, and existing tools are not capable of these nuances, nor does asking AI to produce them help you learn. Your text answers must be your own. Your calculations must also be your own (you may use a calculator or R, but not AI).

AI tools may occasionally be useful to help debug R code, but you should be aware that they regularly produce code that runs but results in incorrect answers. Another permitted use is to translate handwritten equations into typed equations. If you ask AI to help explain a concept for your own understanding, know that it may be wrong (I have seen it happen many times in this area). Always double check everything.

Belonging at Bouvé College:

We acknowledge that chronic and persistent disparities in education and health due to systemic oppression, based on race, class, and other marginalized identities, have adversely affected individuals and society. At Bouvé we are committed to educating future healthcare leaders who strive to serve marginalized communities and improve the well-being of all members of society through the health span.

Within the college, we strive to create a safe environment for diverse thoughts, perspectives, and experiences to be shared, explored, and embraced. As a community we recognize that disrupting long-standing oppressive practices is a process. Everyone is at a different place on this journey, and we affirm our commitment to support and learn from one another.

In order for everyone in our learning community to grow, gain a deeper understanding of complex issues and be exposed to diversity of thought, we may have discussions in this class on topics that could be uncomfortable or difficult for some people. It is imperative that we follow rules of engagement as below in order to create a psychologically safe learning environment.

Ground Rules:

1. Listen actively and respect others when they are talking.
2. Participate to the fullest of your ability without taking up too much space – community growth depends on the inclusion of every individual voice
3. Speak from your own experience instead of generalizing ("I" instead of "we")
4. Do not be afraid to respectfully challenge one another by asking questions but refrain from personal attacks and stay focused on ideas rather than individuals
5. Be conscious of body language and nonverbal responses; they can be as disrespectful and hurtful as words.

Finally, all community members need to adhere to the NU **Student Code of Conduct** (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Fhandbook%2Fcode-student-conduct%2F%23text&data=05%7C02%7CS.Kiami%40northeastern.edu%7C33a01720f98749f1487508dcd27f06bc%7Ca8eec281aaa34daeac9b9a "*each Northeastern student has the responsibility to respect the differences of individuals and treat others in a civil and respectful manner.*"

Belonging at Northeastern (https://belonging.northeastern.edu/)

Students with Disabilities:

Disability Access Services (DAS), located on campus in 10 Dodge Hall (617-373-2675) can provide students with information and other assistance to help manage any challenges that may affect their performance in the coursework. The University requires that students provide documentation of their disability to the DAS. Students should meet with the course instructor for special accommodations to be arranged.

Northeastern University abides by Section 504 of the Rehabilitation Act of 1973, which stipulates that no student shall be denied the benefits of an education "solely by reason of a handicap." Disabilities covered by law include but are not limited to, learning disabilities and hearing, sight or mobility impairments. Additional information about DAS can be found online at **https://drc.sites.northeastern.edu/** (https://drc.sites.northeastern.edu/)

Student Wellness:

As health researchers we do our best to maintain a healthy, balanced life. As a student however, you may experience a range of challenges including significant stress, difficult life events, mood changes, excessive worry, or problems with eating and/or sleeping. These can diminish your academic performance and/or reduce your ability to participate in daily activities. If you or anyone you know is struggling, we strongly encourage you to seek support. Northeastern University provides several services and resources to support the overall wellness of students:

University Health and Counseling Services (https://nam05.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.northeastern.edu%2Fuhcs%2F&data=02%7C01%7Cv.wallace%40northeastern.edu%7C5176bb0a062f4d9b39d108d7

Find at Northeastern (https://nam05.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.northeastern.edu%2Fuhcs%2Ffind-at-northeastern%2F&data=02%7C01%7Cv.wallace%40northeastern.edu%7C5176bb0a062f4d9b39d108d7a36a396c%7Ca8eec281aaa34daeac9b9a3

We Care (https://northeastern.sharepoint.com/:u:/r/sites/BouveCurrentStudentResources/SitePages/We-Care.aspx?csf=1&web=1&e=MQxRpW)

Wellness Days (https://northeastern.sharepoint.com/:u:/r/sites/BouveCurrentStudentResources/SitePages/Wellness-Days.aspx?csf=1&web=1&e=u3zoRZ)

As always, please contact **Northeastern University Police Department** (https://nam05.safelinks.protection.outlook.com/?url=https%3A%2F%2Fnupd.northeastern.edu%2F&data=02%7C01%7Cv.wallace%40northeastern.edu%7C5176bb0a062f4d9b39d108d7a36a396c at 617-373-3333 in emergency situations.

Title IX Policy:

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including students, faculty and staff of all gender identities.

Learn more at:

https://www.northeastern.edu/ouec/training-and-education/syllabus-language-title-ix/ (https://www.northeastern.edu/ouec/training-and-education/syllabus-language-title-ix/)

Know Your Options: Pregnancy (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.flipsnack.com%2FCF5C7988B7A%2Fknow-your-options-pregnancy%2Ffull-view.html&data=05%7C02%7Cg.buttriss%40northeastern.edu%7C88f9f3fa7fac493af13f08dcb5552ae6%7Ca8eec281aaa34daeac9b9a398b9215e7

Course Feedback:

The Office of the Registrar, in collaboration with the Office of the Provost, administers the TRACE (Teacher Rating and Course Evaluation) for undergraduate and graduate courses at Northeastern University. Students' assessments through TRACE are completely anonymous. Given the public nature of student comments made through TRACE and their intended use to enhance teaching, students are asked to please be thoughtful, professional and considerate with responses.

Syllabus Disclaimer:

Any changes to the syllabus will be made at the discretion of the instructor. It is the responsibility of the student to stay informed of any such changes.

For additional information, resources, and policies please visit the **Northeastern University Catalog (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a08dcef9606a** * including:

- **University Undergraduate and University-wide academic policies and procedures (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Fundergraduate%2Facademic-policies-procedures%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a08dcef9606af%7Ca8eec281aaa34daeac9b9a3**
- **University Graduate and University-wide academic policies and procedures (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Fgraduate%2Facademic-policies-procedures%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a08dcef9606af%7Ca8eec281aaa34daeac9b9a3**
- Bouvé College of Health Sciences **Undergraduate (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Fundergraduate%2Fhealth-sciences%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a08dcef9606af%7Ca8eec281aaa34daeac9b9a398** and **Graduate (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Fgraduate%2Fhealth-sciences%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a08dcef9606af%7Ca8eec281aaa34daeac9b9a398** policies, and **program-specific policies and procedures (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Fgraduate%2Fhealth-sciences%2Facademic-policies-procedures%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a08dcef9606af%7Ca8eec281aaa34daeac9b9a3**

Students should consult **the appropriate version based on enrollment (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fcatalog.northeastern.edu%2Farchive%2F&data=05%7C02%7Cv.wallace%40northeastern.edu%7Cf94fc1e0303248f7258a0 of the Northeastern academic catalogs, their academic advisors, and/or their degree audit reports for their specific requirements.*

Major Assignment / Course Schedule

Week	Date	Topics/Assignments/Assessments/etc.
1	Jan 7	Introduction to causal inference
2	Jan 14	Potential outcomes and the ATE
3	Jan 21	Causal diagrams
4	Jan 28	DAGs to distributions
5	Feb 4	Structures of bias
6	Feb 11	Regression adjustment, standardization, and the g-formula
7	Feb 18	Regression adjustment, continued, and causal survival analysis
8	Feb 25	Propensity score methods
9	Mar 4	SPRING BREAK
10	Mar 11	Propensity score methods, continued, and double robustness
11	Mar 18	Bias analysis
12	Mar 25	Interaction, effect heterogeneity, and generalizability
13	Apr 1	Mediation and effect decomposition
14	Apr 8	Quasi-experimental methods
15	Apr 15	Causal inference for complex exposures
16	Apr 22	Final project presentations